SVKM's NMIMS · Mukesh Patel School of Technology Management & Engineering, Mumbai

Modern Day Robotics & Its Industrial Applications (MDRIIA - 702CO0E012) · B.Tech Computer Science & Business Systems (CSBS), in partnership with TCS · AY 2026–27 Odd Semester

GROUP 08 TECHNICAL DOSSIER

Capstone ICA (20 marks): CDIO Robotics Engineering & MuJoCo Physics 10 · CSBS Techno-Economic Business Systems Analysis 5 · Studio Showcase & Viva Voce 5

Group · project	Group 08 · Vision-Guided Multirotor UAV for Last-Mile Flood Disaster Relief Delivery (SkyHydro)
Monorepo folder	https://github.com/sunny-nanade/NMIMS-MPSTME-MDRIIA-2026/tree/main/Group_08_Manikya_Shourya_Keswani_Vora
Submit (3 places)	(1) PDF on MS Teams, turned in by EVERY member · (2) this filled Word file AND its PDF committed to Group_08_Manikya_Shourya_Keswani_Vora/docs/MDRIIA_Group_08_Technical_Dossier.docx / .pdf · (3) LinkedIn video URL in Section 5. Due Monday, 26-Oct-2026, 23:59 IST. Studio Showcase, Expo & Viva Voce: Saturday, 31-Oct-2026.
No slides	There is no PPT for MDRIIA. At the Expo and in your video you present the live MuJoCo simulation, your business sensitivity graphs, and a webcam picture-in-picture of every member.
How to fill this file	Use Microsoft Word (desktop). Choose answers from the 'Select ▾' drop-downs and type into the 'type here' fields. Do not delete or retype them and do not edit this file in Google Docs: the fields are how your answers are read.

Student name	Roll No.	SAP ID	Capstone role(s)	Specialisation (finalised titles sheet / monorepo)
Manikya Rathore	E056	70362400057	Lead Robotics Systems Architect	Lead UAV Aerodynamics, Payload Physics & MuJoCo Modeler
Shourya Garg	E020	70362400047	Autonomous Navigation / Vision Specialist	Computer Vision, Thermal Survivor Detection & Winch Drop Specialist
Keswani Laksh	E032	70362400048	Telemetry QA Lead	Flight Path Optimization & Wind Gust Disturbance Control Lead
Vora Jash	E067	70362400015	CSBS Business & Techno-economic Systems Analyst	CSBS Disaster Logistics, Fleet Economics & Cost-Parity Analyst

Roll numbers and SAP IDs are from the official student list. Tell Dr. Nanade by Tuesday, 20-Oct-2026 if any name, ID or role is wrong.

SECTION 1 · Problem Bottleneck, Industry Standards & Aalborg Research Title *Component 1 · CDIO Robotics (10)*

Authorised Aalborg title	To what extent can an autonomous vision-guided multirotor UAV simulated in MuJoCo for payload-range trade-offs optimize last-mile medical relief drop accuracy during NDRF flood operations while establishing fleet utilization payback parity against ground transport?
Original draft topic	UAV for emergency supply delivery in floods (SkyHydro)
Why the title changed	The student team provided an exceptionally detailed and fact-rich submission, citing genuine Central Water Commission (CWC) and National Disaster Response Force (NDRF) statistics. However, it violated critical academic and institutional guidelines: it embedded raw currency amounts (specific rupee figures per trip) and excessive narrative exposition in the title itself.

1.1 Baseline friction: claim verification table

Each statement was carried over from your titles sheet or your monorepo scripts. Before you use it anywhere (dossier, video, paper), find the primary source yourself and record it, or correct or delete the claim. Statistics: cite the original study, not a blog or an AI answer.

Claim (as carried over)	Faculty pre-audit note	Primary source you verified (author, year, venue, DOI/URL)	Source type	Keep / Correct / Delete
High fleet utilisation of 6–8 round-trip sorties per drone daily during disaster windows.	*An operating assumption. Source it from an NDRF / state disaster-management report or label it as an assumption in your sensitivity graph.*	No primary NDRF/SDMA source found for 6–8 sorties/drone/day. Replaced by our computed cycle: 31 min per sortie (5 km corridor, 8 m/s cruise, sim-measured drop-site time, 10 min battery swap) → 23 sorties/UAV in a 12 h window; an assumption-driven output varied in the sensitivity graph.	No source found	Correct
An 85% reduction in emergency delivery transit latency.	*Compute it from your simulation against a modelled boat baseline, or delete.*	No source found for '85 %'. Computed by our model (analytics/disaster_relief_logistics.py): response latency UAV 34 min vs boat 222 min (M/M/c wait + outbound leg) = 85 % reduction under our labelled assumptions; reported as a model output, not a fact.	No source found	Correct
Central Water Commission and NDRF statistics on submerged highways and boat operations.	*Cite the specific CWC / NDRF document and year for each number you use.*	Central Water Commission (FFM Dte.), 'State-wise flood damage statistics, 1953–2016', letter dt. 10-05-2018, cwc.gov.in/sites/default/files/statewisefloodd atadamagestatistics.pdf: yearly average 31.88 million people affected, 12.42 lakh houses damaged, 1,648 lives lost. No NDRF boat-operation figure could be verified, so none is used.	Government / official report	Correct

1.2 Governing industry standards

Decide for each standard whether it governs your system, and name the clause or requirement you design against (cite number, part and year). 'No' is a correct answer when you can say why. The hint is the faculty view; you may disagree with a reason.

Standard	Scope	Faculty hint for your project	Applies?	Clause / requirement you use, or why not
ISO 10218-1/-2	Safety requirements for industrial robots (robot, and robot integration)	*No*	No	Covers industrial robot arms in a factory cell (ISO 10218-1/-2:2011); SkyHydro is an aerial vehicle in open flood terrain.
ISO 3691-4	Driverless industrial trucks / AMRs and their systems: safety requirements	*No*	No	Driverless industrial trucks / AMRs on the ground (ISO 3691-4:2020); not an aircraft.
ANSI/RIA R15.06	US national adoption of ISO 10218 (industrial robot safety)	*No*	No	US adoption of ISO 10218 for industrial robots; neither the robot type nor the jurisdiction applies.
IEC 60601-1	Medical electrical equipment: basic safety and essential performance	*No*	No	The UAV carries packaged medicines but is not medical electrical equipment applied to a patient (IEC 60601-1:2005+A2:2020).
Domain-specific	India Drone Rules, 2021 (payload drops, airspace zones, BVLOS permissions)	*Recommended: check it*	Yes	Drone Rules, 2021 (MoCA, notified 25-08-2021): R.5 'small' class 2–25 kg AUW incl. payload (ours 4.4–6.4 kg); R.19/22 airspace map, green zone ≤ 120 m AGL (we fly ≤ 18 m), red/yellow need permission; R.28 dangerous goods (Li cells, some medicines); R.31 licensed remote pilot. No payload-drop or BVLOS clause, so we winch to 1 m above the roof and assume a R.48 exemption.

1.3 Documented baseline friction (who struggles, with what, how it is measured today)

Baseline metric and its value today (with unit)	Urgent kit sent by a dedicated rescue boat: response latency 222 min and 105 min boat round trip for a 5 km corridor (modelled); scale: 31.88 million people flood-affected per year in India.
Source of the baseline value (author, year, DOI/URL)	Scale: CWC flood damage statistics 1953–2016 (2018), cwc.gov.in/.../statewiseflooddatadamagestatistics.pdf. Boat speed 8 km/h and 30 min launch are labelled assumptions (no verified NDRF value), varied ±30 %.
Who carries the cost today (role / stakeholder)	NDRF/SDRF boat crews (crew-hours per urgent run), district disaster-management authority (fleet cost), stranded households (waiting time).

SECTION 2 · MuJoCo Rigid-Body Physics Modelling & Python Closed-Loop Control *Component 1 · CDIO Robotics (10)*

MuJoCo mechanism (titles sheet)	MuJoCo MJCF Rigid-Body Model with Vision-Guided Multirotor UAV for Last-Mile Flood Disaster Relief Delivery (SkyHydro) Physics & Closed-Loop Control
MJCF model	Group_08_Manikya_Shourya_Keswani_Vora/models/skyhydro_flood_uav.xml
Controller	Group_08_Manikya_Shourya_Keswani_Vora/src/flood_relief_drop_sim.py
Contact & solver settings in the model today	timestep 0.002 s (500 Hz) · friction 0.7 / 0.01 / 0.001 · solref 0.01 1 · solimp 0.9 0.95 0.001 · 4 rotor-thrust motors + winch
Core state machine (FSM)	TAKEOFF → CRUISE → APPROACH_DROP_ZONE → HOVER → WINCH_LOWER → RELEASE → WINCH_UP → RETURN_TO_BASE
Equations of motion & control	Newton-Euler (rigid body): $m\cdot\dot{v} = \Sigma F$, $I\cdot\dot{\omega} + \omega \times (I\cdot\omega) = \Sigma\tau$; multibody form $M(q)\cdot\ddot{q} + C(q,\dot{q})\cdot\dot{q} + g(q) = \tau + J^{T}\cdot f_{ext}$ Quadrotor Newton-Euler: $m\cdot\ddot{p} = R\cdot[0, 0, \Sigma T_i]^{T} - m\cdot g\cdot e_z + F_{wind}$ Payload pendulum: $\ddot{\theta} = -(g / L)\cdot\sin\theta - (\ddot{x} / L)\cdot\cos\theta$ (L = 3.6 m cable); CEP = radius containing 50% of drop points PID tracking: $u(t) = K_p\cdot e(t) + K_i\cdot\int e\,dt + K_d\cdot\dot{e}(t)$ Damped least-squares IK (manipulators): $\Delta q = J^{T}\cdot(J\cdot J^{T} + \lambda^{2}\cdot I)^{-1}\cdot\Delta x$

Faculty code audit: Keep every cost dimensionless (ratios to the boat baseline); no rupee amounts in the title, abstract or video.

2.1 Model and controller settings you actually used (from your final code)

MJCF timestep (s) in the final model	0.002 s (500 Hz), RK4; controller and logger also run every step (500 Hz)
Main contact friction (sliding / torsional / rolling)	0.7 / 0.01 / 0.001 (rubberised canister base on a concrete roof)
solref (timeconst dampratio) and why you chose it	0.01 1: 10 ms time constant = 5 timesteps (MuJoCo needs ≥ 2·dt) and damping ratio 1, so the canister lands without bouncing; at 0.005 it nears the 2·dt limit and the landing jitters.
Controller law used	PID
Final gains (e.g. Kp, Ki, Kd or λ) with units	Cascaded PID: position Kp 1.2/1.2/1.5 s^{-1}, Ki 0.35/0.35/0.4 s^{-2} (proposed only); velocity Kv 2.5/2.5/3.0 s^{-1}; attitude kR 7.0 N·m/rad, kω 1.1 N·m·s/rad; anti-sway 0.6 (–); canister centring 0.35 s^{-1}; tilt ≤ 35°; winch servo kp 600 N/m, kv 80 N·s/m
FSM states implemented in code (list)	TAKEOFF, CRUISE, APPROACH_DROP_ZONE, HOVER, WINCH_LOWER, RELEASE, WINCH_UP, RETURN_TO_BASE, ABORT (no steady window / battery reserve < 25 %)
Model loads and runs: python -c "import mujoco; mujoco.MjModel.from_xml_path(...)"	Yes

2.2 Authorship of each script

Several scripts were provided as starter scaffolds. Mark honestly what you wrote, changed, or used as given. Viva questions target the code you claim.

File	Written / Modified / Used as given	Main author (Roll)	What you changed or added (1 line)
models/skyhydro_flood_uav.xml	Modified by us	E056	Fixed invalid <motor body>; 4 rotor sites (differential thrust + drag torque), winch slide + releasable canister, roof target
src/flood_relief_drop_sim.py	Written by us	Shared	Replaced the kinematic toy with MuJoCo closed loop: FSM, cascaded PID, wind/gusts, vision release, Monte Carlo, 500 Hz log
analytics/disaster_relief_logistics.py	Written by us	E067	Hard-coded outputs replaced by computed H, κ/NCRI, P_payback, M/M/c, FMEA, ±30 % sensitivity from sim + labelled inputs
analytics/analyze_trials.py	Written by us	E032	Welch t-test, Cohen's d, 95 % CI, CEP, box + trajectory plots from *_real_trials.csv
simulate.py, environment/, rl/	Written by us	E056	3D valley, animated water, PPO pilot (stages 1–3): stage 3 completed 30/30 missions, 150 s vs 309 s expert (rl_pilot_real_trials.csv)

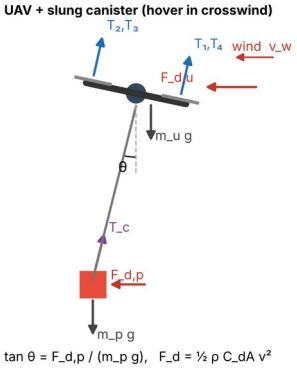
2.3 Key code snippet (≤ 25 lines) implementing your FSM or control law:

src/flood_relief_drop_sim.py, lines 262–286 (FSM transitions: HOVER → WINCH_LOWER → RELEASE / ABORT)

```
 262            elif s == "HOVER":
 263                if self.ctrl_name == "baseline":
 264                    ok = err < 0.5 and np.linalg.norm(v) < 0.5
 265                else:
 266                    ok = np.linalg.norm(self.est_err) < 0.6 and np.linalg.norm(v) < 0.5
 267                if (ok and ts > 1.0) or ts > 15.0:
 268                    self.set_state("WINCH_LOWER")
 269            elif s == "WINCH_LOWER":
 270                if self.d.qpos[self.q_winch] >= self.winch_target - 0.03:
 271                    self.set_state("RELEASE")
 272            elif s == "RELEASE":
 273                if self.ctrl_name == "baseline":
 274                    go = True                                          # timed release at the waypoint
 275                else:
 276                    go = (np.linalg.norm(self.est_err) < REL_ERR and np.linalg.norm(self.est_rel_v) <
REL_SPD)
 277                if go:
 278                    self._release()
 279                    self.set_state("WINCH_UP")
 280                elif ts > WINDOW_TIMEOUT:
 281                    self.outcome = "aborted_no_steady_window"
 282                    self.set_state("ABORT")
 283            elif s in ("WINCH_UP", "ABORT"):
 284                if self.d.qpos[self.q_winch] <= -3.2 or ts > 12.0:
 285                    self.set_state("RETURN_TO_BASE")
 286                    self._new_segment(p, np.array([PAD[0], PAD[1], Z_CRUISE]), V_CRUISE)
```

2.4 Free-body diagram and one equation worked with your own numbers and units:

Free-body diagram (hover in crosswind, docs/figures/fig_fbd.png) and worked example



Symbols: T_1…T_4 rotor thrusts (N, ≤ 35 N each); m_u = 3.355 kg airframe + cable + hook; m_p canister mass (kg); $F_d = \tfrac{1}{2}\rho C_{dA} v^2$ quadratic drag, ρ = 1.225 kg/m³, C_{dA} = 0.10 m² (UAV) and 0.08 m² (canister: 0.28 × 0.36 m, C_d 0.8); θ cable swing angle; T_c cable tension; φ airframe tilt.

Worked example (hover, U = 10 m/s, m_p = 2 kg): $F_{d,p} = \tfrac{1}{2}\cdot 1.225\cdot 0.08\cdot 10^2 = 4.90$ N; $\tan\theta = F_{d,p}/(m_p g) = 4.90/19.62 = 0.250 \rightarrow \theta = 14.0°$. Vertical: $\Sigma T\cdot\cos\varphi = (m_u + m_p) g = 52.5$ N → 13.1 N per rotor. Horizontal: $F_{d,u} + F_{d,p} = 6.13 + 4.90 = 11.03$ N → tilt $\varphi = \text{atan}(11.03/52.5) = 11.9°$.

In our telemetry: run 180 (run_180_proposed_w10.csv.gz), m_p = 2.66 kg, HOVER state, 501 samples at 500 Hz: swing_deg = 9.7 ± 1.2° vs 9.8° predicted for the mean wind 9.6 m/s (gusts make it swing about that value); mean T1_N…T4_N = 16.1 N per rotor vs 14.7 N static prediction.

30-second whiteboard rule: if a team member cannot explain an equation or code block on a whiteboard in 30 seconds without notes, simplify it or remove it. It will be asked in the viva.

SECTION 3 · CSBS Techno-Economic Business Systems Modelling *Component 2 · CSBS Techno-Economic (5)*

Business script	Group_08_Manikya_Shourya_Keswani_Vora/analytics/disaster_relief_logistics.py (rename or add *_roi.py if you restructure it)
CSBS business case (titles sheet)	CSBS Disaster Logistics & Fleet Utilization Economics: Ground delivery during monsoon floods is obstructed by submerged highways, requiring expensive chartered diesel rescue boats with limited payload and long transit times. Comparative operational cost-parity analysis reveals that while initial UAV hardware procurement constitutes capital investment, high fleet utilization (operating 6-8 round-trip sorties per drone daily during disaster windows) achieves operational cost parity against ground/water transport within the first emergency relief season. Crucially, the 85% reduction in emergency delivery transit latency prevents preventable mortality among stranded populations.
Constants hard-coded in the script today	mean_transit_distance_km = 8.5, v_boat_kmh = 7.2, t_boat_launch_prep_min = 35.0, v_uav_kmh = 45.0, c_uav_battery_wear = 0.15, k_capex_fleet = 0.45 → each one needs a source in 3.1 or must be computed from your simulation.
Owner	CSBS Business & Techno-economic Systems Analyst: E067. Tags in this section carry that roll number.

Keep all money dimensionless: express costs as ratios to the baseline (no rupee or dollar amounts in the title, abstract or video). Every input needs a source or must be a clearly labelled assumption that you vary in the sensitivity graph.

3.1 Business model: formulas, your inputs and your results

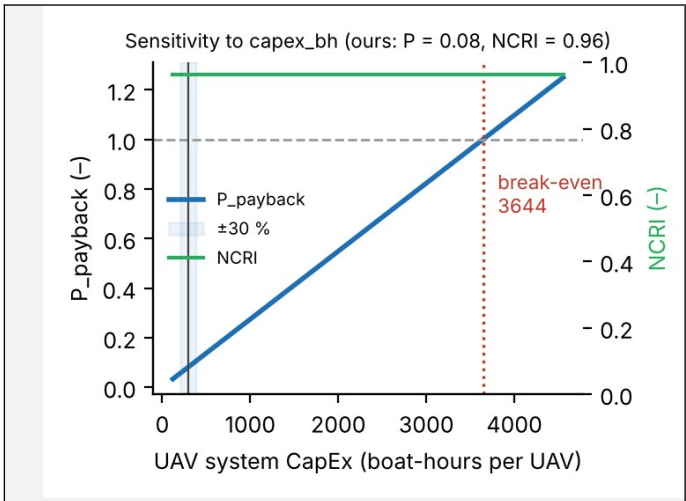
Quantity	Formula (define every symbol in your script)	Inputs used, with source or 'assumption'	Result (with unit)
Labour reallocation H_reclaimed	H = boat-crew hours per day replaced by UAV sorties (crew-hours/day)	λ = 2 requests/h × 12 h/day [assumption]; boat round trip 105 min [assumption: 5 km, 8 km/h, 30 min launch]; crew 4 [assumption]	168 boat-crew hours/day (24 urgent deliveries/day moved to UAVs)
Payback horizon P_payback (dimensionless)	P_payback = CapEx / (T_life · ΔOpEx_annual); P < 1 ⇒ pays back within life T_life. Also report T_payback = CapEx / ΔOpEx in months.	CapEx = 300 boat-h per UAV × 2 [assumption]; ΔOpEx from κ below × 24/day × 60 flood days/yr [assumption]; T_life 3 yr [assumption]	P_payback = 0.082 (–); T_payback = 3.0 months
Normalised Cost Reduction Index NCRI	NCRI = (C_baseline − C_robot) / C_baseline (0–1); cost-parity ratio κ = C_robot / C_baseline = 1 − NCRI	C_boat = round trip × 1 boat-h; C_uav = (1/p_success) × flight time × r_uav; p_success 0.833 and drop-site time 10 s [SIM]; r_uav 0.15 [assumption]	κ = 0.036; NCRI = 0.964 (–)
Queuing delay	M/M/c: relief requests (λ per hour) served by c UAVs (μ sorties per hour each). ρ = λ / (c·μ) < 1; M/M/1: W_q = ρ / (μ − λ); M/M/c: W_q = C(c, λ/μ) / (c·μ − λ) (Erlang C)	λ = 2/h [assumption]; μ_UAV = 1.61/h from cycle 31 min incl. re-flights [SIM+assumption]; c = 2 UAVs vs 4 boats (μ = 0.57/h)	UAV ρ = 0.62, W_q = 23.2 min; boats ρ = 0.88, W_q = 155 min
Sensitivity analysis	Vary the most uncertain input ±30 % (or across your scenario range) and plot P_payback and NCRI; report the break-even value.	All assumptions ±30 % (tornado); UAV CapEx = most uncertain input (no verified price data), swept 0.4× to 15× in the graph	Most influential (tornado): λ, break-even P = 1 at λ = 0.16 requests/h; CapEx break-even P = 1 at 3,644 boat-h/UAV (12× assumed); NCRI = 0 at r_uav = 4.2

Queuing model used	M/M/c
λ, μ, c values used (with units)	λ = 2 requests/h; μ = 1.61 sorties/h per UAV; c = 2 UAVs (boats: μ = 0.57/h, c = 4)
Payback in months (T_payback)	3.0 months (calendar; savings accrue only in the 60 flood days/yr)
Script output reproduces every number above when re-run	Yes

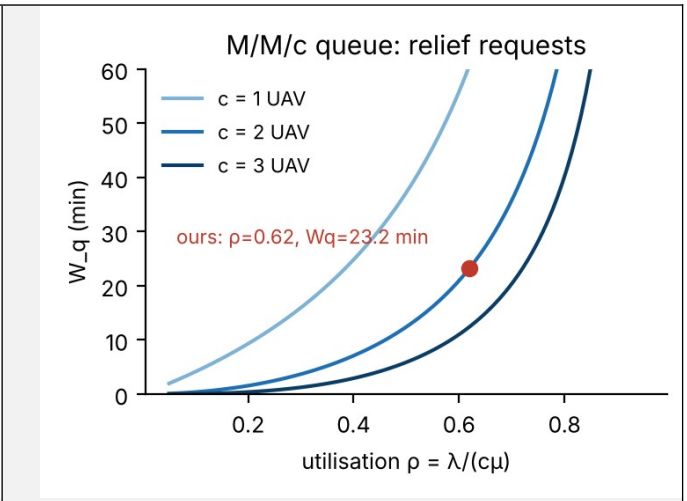
3.2 FMEA: failure modes, effects and risk priority (RPN = S × O × D, each 1–10)

Two failure modes are seeded from your project; add at least two more. Score before and after your mitigation (design change, control limit, sensor, procedure). The post-mitigation RPN must be justified by something you can show in the simulation or the code.

Failure mode	Effect on mission / business	S	O	D	RPN pre	Mitigation (what you changed)	RPN post	Shown in
Payload swing in 15 m/s crosswind → drop outside target	kit lands in water / wrong roof; re-fly, medicine lost	8	7	5	280	vision canister centring + anti-sway + steady release window; ABORT instead of a blind drop	48	Both
Battery depletion before return to base	UAV ditched in flood water; fleet capacity lost	9	3	6	162	power model in the loop; ABORT/RTB if predicted SoC at the pad < 25 % (--battery-soc 0.3 demo)	36	Code
Canister released while still swinging (timed release)	drop error grows with swing; kit damaged	7	6	6	252	release only if canister error < 0.35 m and rel. speed < 0.35 m/s, else wait / abort	28	Both
Vision noise / glare on the drop marker	wrong centring → off-zone drop	6	4	6	144	α-β filter on 30 Hz camera data; 5 cm noise in every trial	54	Simulation
Rotor thrust saturation in gusts	position loss near buildings, possible crash	9	3	5	135	35° tilt limit, T/W ≥ 2.2 at 3 kg, integral wind rejection; crash check every run	54	Both



P_payback and NCRI vs UAV CapEx (most uncertain input), break-even marked



M/M/c waiting time vs utilisation, our operating point

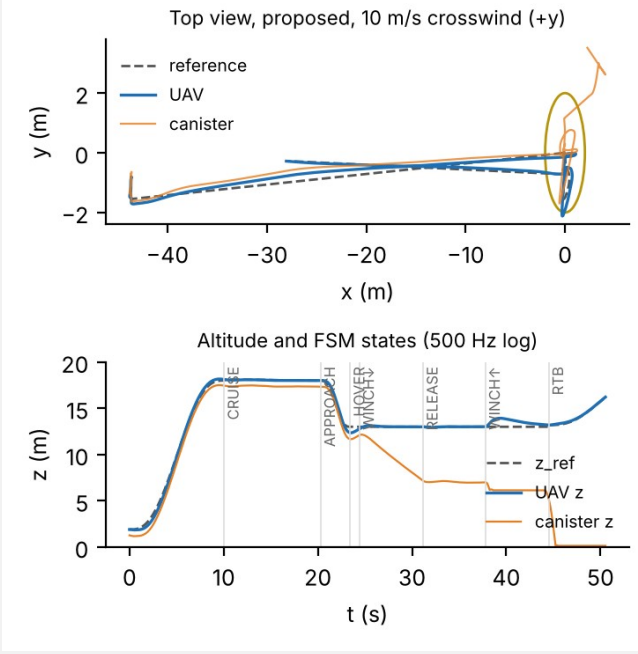
SECTION 4 · Simulation Telemetry, Monte Carlo Testing & Empirical Validation *Component 1 · CDIO Robotics (10)*

Evaluation protocol (titles sheet)	Conduct N = 40 simulated drops across four crosswind velocities (0, 5, 10, 15 m/s) with varying payload weights (1.0 to 3.0 kg). Measure: Circular Error Probable (CEP in meters), payload swing angle (degrees), flight duration to battery depletion, and delivery success rate.
Telemetry standard	Log at 500 Hz (MJCF timestep 0.002 s, or every 2nd step at 0.001 s): time, state (q, q̇ or pose, velocity), reference, tracking error, actuator torques/forces, FSM state, events. One CSV per run or one CSV with a run_id column.
Monte Carlo	N ≥ 30 runs per condition (more if your protocol says so). Randomise disturbances and initial conditions from stated distributions; record every random seed.
Required measures	Tracking RMSE = $\sqrt{(\Sigma\|p - p_{ref}\|^2 / n)}$ on the planned flight path to the drop zone vs MuJoCo UAV position; control effort E = $\int \tau^2 dt$ (sum of $\tau^2 \cdot \Delta t$ over all actuators); your primary task metric from the protocol.
Statistics	Compare the proposed controller with a baseline condition: Student's / Welch's t-test (two-sided, α = 0.05), Cohen's d = $(\bar{x}_1 - \bar{x}_2)$ / s_pooled, and a 95 % confidence interval.
Where real trial data goes	Group_08_Manikya_Shourya_Keswani_Vora/analytics/*_real_trials.csv, produced by your controller. Files named SIMULATED_* are starter targets, never results.
Owner	Telemetry QA Lead: E032.

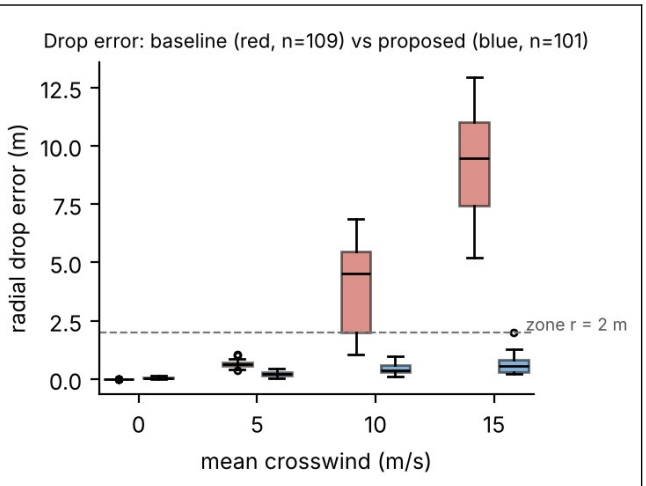
Baseline condition you compare against	'GPS waypoint drop': same airframe and cascaded PD loop on the DRONE position, no canister sensing, no integral, canister released as soon as the winch is down
Monte Carlo runs per condition (N)	30 per crosswind × 4 crosswinds = 120 per controller (240 total)
Randomised parameters and their distributions	Crosswind U ∈ {0, 5, 10, 15} m/s + Gauss–Markov gusts (σ = 0.1U, τ = 2 s); canister mass ~ U(1, 3) kg; launch point ~ U(±2 m); camera noise N(0, 5 cm). Same seed for both controllers.
Logging rate actually achieved (Hz)	500 Hz (every 0.002 s step; mean 28882 samples/run)
Random seeds recorded for every run	Yes
Real-trials CSV path(s) in the repo	Group_08_Manikya_Shourya_Keswani_Vora/analytics/ flood_relief_real_trials.csv; 500 Hz logs: analytics/telemetry/*.csv.gz

Measure	N	Baseline mean ± SD	Proposed mean ± SD	Unit	Better? (direction)
Tracking RMSE	120 / 120	0.953 ± 0.419	0.700 ± 0.236	m	Lower is better
Control effort $\int \tau^2 dt$	120 / 120	47877 ± 39130	32993 ± 11093	$N^2 \cdot s$ (Σ rotor T^2 dt)	Lower is better
Radial drop error (released runs)	109 / 101	2.820 ± 3.553	0.306 ± 0.323	m	Lower is better

Test used	Welch's t-test
t statistic, degrees of freedom, p-value (primary metric)	t = 7.35, df = 109.9, p = 3.7e-11 (two-sided; Mann–Whitney p = 3.1e-06)
Cohen's d (primary metric)	d = 0.98
95 % confidence interval of the difference	[1.837, 3.192] m (baseline – proposed)



Representative run (proposed, 10 m/s): reference vs actual, FSM states



Radial drop error by crosswind, baseline vs proposed (n = 109 / 101)

DATA PROVENANCE DECLARATION (mandatory). Reporting generated target numbers as experimental results is research misconduct and scores zero for this component.

Every number in Section 4 comes from MuJoCo runs of OUR controller, logged by our code	Yes
No value comes from SIMULATED_* files, generate_paper_figures.py or any np.random 'target' data	Yes
Our plots re-generate from the *_real_trials.csv files in the repo	Yes
Every input in Section 3 has a source or is labelled as an assumption	Yes

SECTION 5 · LinkedIn Video Showcase, AI Verification Audit & Peer Review *Component 3 · Showcase & Viva (5)*

LinkedIn post URL	type here
Video length (mm:ss)	type here
Video shows the live MuJoCo simulation (not slides)	Select ▾
Video shows our business sensitivity graph	Select ▾
Webcam picture-in-picture: every member appears and speaks	Select ▾

5.1 AI Interaction Audit Log: three cases where AI gave wrong robotics or economic data

Examples: a wrong MuJoCo attribute or API call, an invented statistic, a correct paper given the wrong journal, an economic formula with the wrong units. Show how you checked each one with a textbook, the official documentation or the original paper.

#	AI tool	What we asked & what AI said	How we caught it	What was actually true (with source)	Error type	Verdict
AI-01	Claude	Claude's water-texture baker started one worker process per CPU thread (23) 'to use the whole lab PC'.	Ran / tested the code or simulation	Setup crashed on the 32 GB lab PC ('OpenBLAS: memory allocation failed', BrokenProcessPool): each worker opened 24 BLAS threads. Fixed: ≤ 8 workers, OPENBLAS_NUM_THREADS = 1 (0.6 GB/worker measured).	Wrong code / MuJoCo API	Corrected
AI-02	Claude	Claude claimed GPU training would speed PPO up and estimated 3.5–6 h for the curriculum, '1 M steps in 10–15 min on 16 cores'.	Ran / tested the code or simulation	GPU stayed ~20 % (bursts to 100 % only in updates); SB3 warns MLP-PPO is CPU-bound (SB3 issue #1245). Measured ~490 steps/s with 22 CPU envs: stage 3 (3 M steps) took 104 min. MuJoCo on the CPU sets the speed.	Wrong number or statistic	Corrected
AI-03	Claude	Claude's first canister-centring law set the drone target = marker − (low-passed canister offset) to put the canister over the marker.	Ran / tested the code or simulation	At 15 m/s the loop fed the swing back into the reference and oscillated (offset grew to 9 m, release never possible). Replaced by a slow integral shift on the canister error (0.35 s^{-1}) + anti-sway; checked in the viewer.	Wrong physics / equation	Corrected

5.2 Peer review: watch both assigned LinkedIn videos and score them (0–2 per criterion)

Group reviewed	Problem	Physics	Business	Data	Clarity	Best moment (mm:ss)	One question for the viva
Group 09: Autonomous Ground Vehicle (UGV) with LiDAR Perception for Hazardous Terrain Reconnaissance	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	type here	type here
Group 02: AI Companion Robot for Remote Elderly Supervision & Fall Emergency Response	Select ▾	Select ▾	Select ▾	Select ▾	Select ▾	type here	type here

SECTION 6 · Individual Learning Record & Research Consent *unmarked · one row per member · answer honestly*

These answers do not change your marks. They help improve the course and are used in anonymised research only if you consent in the sign-off below. Commit share: your commits ÷ the group's commits in your folder (git shortlog -sn -- <folder>).

Member	Hours I spent (total)	My declared role matched my actual work (1 = not at all, 5 = fully)	I learned a skill outside my role (robotics ↔ business)	Scripts I mainly wrote (file names)	My commit share (%)	Share of my writing first drafted with AI	Wrong AI facts / code I personally caught
Manikya Rathore (E056)	type here	Select ▾	Select ▾	type here	type here	Select ▾	Select ▾
Shourya Garg (E020)	type here	Select ▾	Select ▾	type here	type here	Select ▾	Select ▾
Keswani Laksh (E032)	type here	Select ▾	Select ▾	type here	type here	Select ▾	Select ▾
Vora Jash (E067)	type here	Select ▾	Select ▾	type here	type here	Select ▾	Select ▾

Member	Before this course I had used a physics simulator	The most valuable thing I learned (one line)	One thing I would change in this course (one line)
Manikya Rathore (E056)	Select ▾	type here	type here
Shourya Garg (E020)	Select ▾	type here	type here
Keswani Laksh (E032)	Select ▾	type here	type here
Vora Jash (E067)	Select ▾	type here	type here

Track A: how this dossier becomes your 4-page IEEE paper

I. Introduction	Section 1: title, verified baseline friction, governing standards
II. Related work	Section 1.1: verified sources only (5+ with DOIs)
III. System model	Section 2: MJCF model, FSM, equations, control law
IV. Techno-economics & evaluation	Sections 3–4: CSBS model, FMEA, Monte Carlo results, statistics
V. Discussion & limitations	Sim-to-real gap, assumptions in the business model, AI audit lessons
Target venues (titles sheet)	A: IEEE INDICON · B: IEEE ICRA

Individual Academic Sign-off *every member answers and signs*

Answer each statement for yourself: (1) this work is original and every source and AI use is declared; (2) our model loads and our controller and business scripts run from the repository; (3) I understand our whole project and can explain my own part in the viva voce.

Member (Roll)	(1) Original work	(2) Code runs	(3) Viva-ready	Signature	Optional research consent*
Manikya Rathore (E056)	Select ▾	Select ▾	Select ▾		Select ▾
Shourya Garg (E020)	Select ▾	Select ▾	Select ▾		Select ▾
Keswani Laksh (E032)	Select ▾	Select ▾	Select ▾		Select ▾
Vora Jash (E067)	Select ▾	Select ▾	Select ▾		Select ▾

**Optional: may Dr. Nanade use anonymised data from this dossier and repository in education research publications? Your answer has no effect on your marks, and you can withdraw until results are published.*